MH1812 Discrete Mathematics
- Continuous Assessment 1

NTU, AY19/20 S1 — 19 Sept 2019, 10:30AM - 11:20AM

Name: Tutorial Group: NTU Email:

Note: ***first 3*** *questions are compulsory. Question 4 is bonus, you get the points if you answer it correctly, you don't lose any points if you leave it or answer wrongly. You can write on the back of paper if there is not enough space, leave the tutorial group if you can't remember.*

Question 1 (30 points)

1. Compute $2019 + 2020 + 2021 + 2022$ modulo 4. (10 points)

   Solution: Since $2019+2020+2021+2022 = 2019+(2019+1)+(2019+2)+(2019+3)$, $2019 + 2020 + 2021 + 2022 \equiv 4 \times 2019 + (1 + 2 + 3) \equiv 6 \equiv 2 \mod 4$.

   Alternatively, $2019 \equiv 4 \times 504 + 3 \equiv 3 \mod 4$
   $2020 \equiv 4 \times 505 + 0 \equiv 0 \mod 4$
   $2021 \equiv 4 \times 505 + 1 \equiv 1 \mod 4$
   $2022 \equiv 4 \times 505 + 2 \equiv 2 \mod 4$
   Therefore $2019 + 2020 + 2021 + 2022 \equiv 3 + 0 + 1 + 2 \equiv 6 \equiv 2 \mod 4$ $\square$

2. Consider the domain $\mathbb{R}$ and suppose that $P(x, y)$ is the predicate $P(x, y) =: x^3 - y^3 \geq 0$. What are the truth values of these statements:

   (a) $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, P(x, y)$. (10 points)

   Solution:The statement is true. If $x \geq 0$, take $y = 0$, then $x^3 - y^3 \geq 0$. If $x < 0$, $y = 2x$, then $x^3 - y^3 = -7x^3 \geq 0$ $\square$

   (b) $\exists x \in \mathbb{R}, \forall y \in \mathbb{R}, P(x, y)$. (10 points)

   Solution:The statement is false. By contradiction, if there is $x_0 \in \mathbb{R}$ such that $\forall y \in \mathbb{R}, P(x, y)$. If $x_0 = 0$, then take $y_0 = 1$. If $x_0 > 0$, take $y_0 = 2x_0$, then $x_0^3 - y_0^3 = -7x_0^3 < 0$. If $x_0 < 0$, take $y_0 = X_0/2$, then $x_0^3 - y_0^3 = 7X_0^3/2 < 0$. $P(x_0, y_0)$ is not true. $\square$

Question 2 (40 points)

1. Show that $(p \wedge (\neg(\neg p \vee q))) \vee (p \wedge q) \equiv p$(20 points).

   Solution:

$$
\begin{aligned}
(p \wedge (\neg(\neg p \vee q))) \vee (p \wedge q) &\equiv (p \wedge (\neg\neg p \wedge \neg q)) \vee (p \wedge q) \quad \text{(De Morgan)} \\
&\equiv (p \wedge (p \wedge \neg q)) \vee (p \wedge q) \quad \text{(Double Negation)} \\
&\equiv ((p \wedge p) \wedge \neg q) \vee (p \wedge q) \quad \text{(Associativity)} \\
&\equiv (p \wedge \neg q) \vee (p \wedge q) \quad \text{(Idempotent)} \\
&\equiv p \wedge (\neg q \vee q) \quad \text{(Distributivity)} \\
&\equiv p \wedge T \quad \text{(Tautology)} \\
&\equiv p
\end{aligned}
$$

□

   Alternatively, one can prove the equivalence by constructing their truth table.

2. Decide whether the following argument is valid (20 points):

$$
\begin{aligned}
&\neg p \to q; \\
&\neg q \to p; \\
&\therefore \neg(p \wedge q)
\end{aligned}
$$

   Solution: We have the following truth table

p	q	$\neg p$	$\neg q$	$\neg p \to q$	$\neg q \to p$	$\neg(p \wedge q$
T	T	F	F	T	T	F
T	F	F	T	T	T	T
F	T	T	F	T	T	T
F	F	T	T	F	F	T

   The first, second and third rows are critical rows, but first row has the false conclusion, therefore the argument is not valid.

   Alternatively, you can make the following arguments. Both premises are equivalent statement $p \vee q$. They are true, so we have three cases: (1) p is true, q false; (2) p is false, p is true; (3) both p and q are true. For the case (3), the conclusion is false. It means that the argument is not valid, and we have the counter-example: $p = T$ and $q = T$.

□

Question 3 (30 points)

Use Mathematical Induction to prove that for every integer $n \geq 1$,

$$2 + 4 + 6 + \cdots + 2n = n^2 + n.$$

Solution:Let $P(n) : 2 + 4 + 6 + \cdots + 2n = n^2 + n$.
$P(1) : 2 \leq 1^2 + 1 = 2$, which is true as both sides are equal to 2. Assume $P(k)$ is true: $2 + 4 + 6 + \cdots + 2k = k^2 + k$, We show that $P(k+1)$ is also true.

$$2+4+6+\cdots+2k+2(k+1) = k^2+k+2(k+1) = k^2+2k+1+k+1 = (k+1)^2+(k+1)$$

This shows that $P(k+1)$ is also true. $\square$

Question 4 (Bonus Question) (10 points)

Prove or disprove: for each integer $n \geq 0, 2^{2n} - 1$ is divisible by 3.

Solution: The statement is true. We prove it by mathematical induction. Let $P(n)$: $2^{2n} - 1$ is divisible by 3, $n \geq 0$.
$P(0)$ is true: as $2^{2\times 0} - 1 = 1 - 1 = 0$ is divisible by 3. Assume $P(k)$ is true: $2^{2k} - 1$ is divisible by 3. We may assume that $2^{2k} - 1 = 3t$, for some integer t. We show that $P(k+1)$ is also true.
$2^{2(k+1)} - 1 = 2^{2k+2} - 1 = 2^{2k} \times 2^2 - 1 = 4 \times 2^{2k} - 1 = 3 \times 2^{2k} + 2^{2k} - 1 = 3 \times 2^{2k} + 3t = 3(2^{2k} + 3)$, which is divisible by 3. This gives the proof.

$\square$